

ML and DS Internship Practical

Month 2 – Intermediate Python

Library Management System Using Python (OPP)

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Submitted To: -

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Technology Used: Python

Development Environment: Visual Studio Code

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1. Project Title: -

Library Management System Using Python (Object-Oriented Programming)

2 Introduction: -

The Library Management System is a console-based application developed using Python. The project is designed to manage library books efficiently through a simple menu-driven interface. It allows users to add, view, search, issue, return, and delete books.

The main objective of this project is to understand and implement Object-Oriented Programming (OOP) concepts such as classes, objects, methods, and encapsulation while developing a real-world application.

3. Objectives: -

The objectives of this project are:

- To understand the fundamentals of Object-Oriented Programming.
- To implement classes and objects in Python.
- To apply encapsulation for data security.
- To create a menu-driven application.
- To perform CRUD (Create, Read, Update, Delete) operations on book records.
- To improve logical thinking and problem-solving skills.

4. Problem Statement: -

Managing library books manually can be time-consuming and error-prone. This project aims to automate basic library operations such as adding books, searching books, issuing books, returning books, and deleting books using Python.

5. Feature: -

- View All Books
- Search Book by Book ID
- Issue Book
- Return Book

- Delete Book
- User-Friendly Menu
- Input Validation (Optional)
- Exit Application

6. Technologies Used: -

- Python 3.x
- Visual Studio Code
- Git
- GitHub

7. OOP Concepts Used: -

Class

A class is a blueprint for creating objects.

Classes used in this project:

- Book
- Library

Object

Objects are instances of a class.

Example:

- book1
- library

Methods

Methods are functions defined inside a class.

Methods used:

Book Class

- get_book_id()
- get_title()
- get_status()
- issue_book()
- return_book()
- display()

Library Class

- add_book()
- view_books()
- search_book()
- issue_book()
- return_book()
- delete_book()

Encapsulation

Encapsulation is used by making book details private using double underscores (__).

Example:

- __book_id
- __title
- __author
- __status

Private data can only be accessed through getter methods.

8. System Requirements: -

Hardware Requirements

- Laptop/Desktop
- Minimum 4 GB RAM
- Intel i3 Processor or Higher

Software Requirements

- Windows 10/11
- Python 3.x
- Visual Studio Code
- Git

9. Project Structure: -

Library-Management-System

|— library_management.py

|— README.md

|— Documentation.pdf

|— Screenshots

|— Testing_Report.pdf

10. Working of the Project: -

Step 1

The program starts by displaying the main menu.

Step 2

The user selects an option.

Step 3

Depending on the selected option, the system performs the required operation.

Examples:

- Add a new book
- View all books
- Search a book
- Issue a book
- Return a book
- Delete a book

Step 4

The menu is displayed again until the user chooses Exit.

11. Algorithm: -

Add Book

1. Read Book ID.
2. Read Book Title.
3. Read Author Name.

4. Create Book object.
5. Store object in Library list.
6. Display success message.

View Books

1. Check if the library is empty.
2. If empty, display a message.
3. Otherwise, display all book details.

Search Book

1. Read Book ID.
2. Compare with stored books.
3. If found, display details.
4. Otherwise, display "Book Not Found."

Issue Book

1. Read Book ID.
2. Search the book.
3. If available, change status to "Issued."
4. Display confirmation.

Return Book

1. Read Book ID.
2. Search the book.
3. Change status to "Available."
4. Display confirmation.

Delete Book

1. Read Book ID.
2. Search the book.
3. Remove the book from the list.

4. Display success message.

12. Advantage: -

- Easy to use
- Simple menu-driven interface
- Reduces manual work
- Demonstrates Object-Oriented Programming
- Easy to modify and extend
- Beginner-friendly project

13. Screenshot: -

```
===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Delete Book
7. Exit
Enter your choice: 1
Enter Book ID: 01
Enter Book Title: Peer e kamil
Enter Author Name: Umera Ahmed
Book added successfully.
```

```
===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Delete Book
7. Exit
Enter your choice: 2

----- Library Books -----

Book ID : 01
Title   : Peer e kamil
Author  : Umera Ahmed
Status  : Available
```

```
===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Delete Book
7. Exit
Enter your choice: 3
Enter Book ID to search: 01

Book Found:

Book ID : 01
Title   : Peer e kamil
Author  : Umera Ahmed
Status  : Available
```



```
===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Delete Book
7. Exit
Enter your choice: 4
Enter Book ID to issue: 01
Book issued successfully.
```

```
===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Delete Book
7. Exit
Enter your choice: 5
Enter Book ID to return: 01
Book returned successfully.
```

```
===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Delete Book
7. Exit
Enter your choice: 6
Enter Book ID to delete: 01
Book deleted successfully.
```

```
===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Delete Book
7. Exit
Enter your choice: 7
Thank you for using the Library Management System.
```

14. Testing Report: -

Test Case ID	Test Scenario	Test Input	Expected Result	Actual Result
TC-01	Add a new book	Book ID: 101, Title: Python Programming, Author: John Smith	Book should be added successfully	Book added successfully
TC-02	View all books	Select View Books option	All stored books should be displayed	Books displayed correctly
TC-03	Search existing book	Book ID: 101	Book details should be displayed	Book found and displayed
TC-04	Search non-existing book	Book ID: 999	Display " Book not found. "	Correct message displayed
TC-05	Issue available book	Book ID: 101	Book status changes to Issued	Book issued successfully
TC-06	Issue already issued book	Book ID: 101 (already issued)	Display " Book is already issued. "	Correct message displayed
TC-07	Return issued book	Book ID: 101	Book status changes to Available	Book returned successfully
TC-08	Return available book	Book ID: 101 (already available)	Display " Book is already available. "	Correct message displayed
TC-09	Delete existing book	Book ID: 101	Book should be removed from the library	Book deleted successfully
TC-10	Delete non-existing book	Book ID: 999	Display " Book not found. "	Correct message displayed
TC-11	View books when library is empty	No books available	Display " No books available. "	Correct message displayed
TC-12	Exit application	Select Exit (7)	Program should terminate successfully	Program exited successfully

15. Learning Outcomes: -

After completing this project, the following concepts were learned:

- Classes and Objects
- Constructors (init)
- Methods
- Encapsulation

- Lists of Objects
- Menu-Driven Programming
- Conditional Statements
- Loops
- Functions
- Problem Solving Using Python

16. Limitations: -

- Data is stored temporarily (lost after program exits if file/database is not used).
- Single-user application.
- Console-based interface.
- No login or authentication system.

17. Conclusion: -

The Library Management System successfully demonstrates the implementation of Object-Oriented Programming concepts in Python. The project provides a simple and efficient way to manage library books through a console-based interface. It improves programming skills by applying classes, objects, methods, and encapsulation in a practical application. This project serves as a strong foundation for developing more advanced library management systems using databases and graphical user interfaces in the future.

18. GitHub Repository Link: -

https://github.com/saniya2698/Library_Management_system_using_OOP